CROSSSAFE · PROJECT PLAN

CrossSafe project plan

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/crosssafe		
Source	docs/11-plan.md @ d9f28be		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	First project plan (kit 1.9.1 upgrade); P4 to P6 shown as planned work only, on hold under the stage cap

CrossSafe project plan

The repo is at stage P3 (TRL 3, constructable design on paper, every option modelled at D3). The next gate is Amish's decision on the concept (O1) and the open items O2 to O4; nothing past TRL 3 is planned for execution while the portfolio cap holds. Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Concept and calculations	Design lead	CRS-CAL-001 v0.7, CRS-STM-001, CRS-OPT-001	Done at P3
CAD at D3 for every option	Design lead	`cad/src/model.py`, `cad/fidelity.json`, STEP files	Done at P3
Requirements and traceability	Design lead	CRS-URS-001, CRS-PRS-001, CRS-RTM-001	Done at P3
Build plan	Design lead	CRS-BLD-001 (paper)	Done at P3
Regulatory, risk, market	Design lead	CRS-REG-001, CRS-RSK-001, CRS-MKT-001	First pass; many rows "to assess"
Decisions	Amish	O1 to O4 in CRS-DEC-001	Awaiting Amish

Resources needed

Table 2. Resources for the planned stages (estimates, to assess; not authorised).

Resource	Deliverable	Effort	Estimated cost (USD)
Pilot road authority contact	Rules, site, permit (P4)	2 weeks elapsed	To assess
Workshop with sheet-metal tools	Prototype build (P4)	3 weeks	364 parts plus labour
Photometry and thermal tests	Intensity, flash area, battery temperature (P4)	2 weeks	To assess
Clinical-safety reviewer	Flash area review (H5)	1 day	To assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P3	Constructable design, calculations, every option at D3, STEM check, options, requirements, registers	This repo, awaiting Amish's review
P4	Prototype build and bench test	On hold (planned only)
P5	Site trial at a pilot crossing	On hold (planned only)
P6	Release and documentation of a tested design	On hold (planned only)

Next steps and decision requested

Amish is asked to decide O1 (concept: A, C or both), O2 (55 C battery limit), O3 (flash area) and O4 (cost over target), all in CRS-DEC-001. The recommended next step is to review them and then ask the portfolio whether CrossSafe is ready for TRL 4; this note does not start it.

References

- FHWA, IA-21 and RRFB countermeasure pages (CRS-PRB-001).
- Carmanah, BLT Direct and Kirkland pages (CRS-MKT-001).
- U.S. Access Board, Chapter 3: Operable Parts (CRS-REG-001).